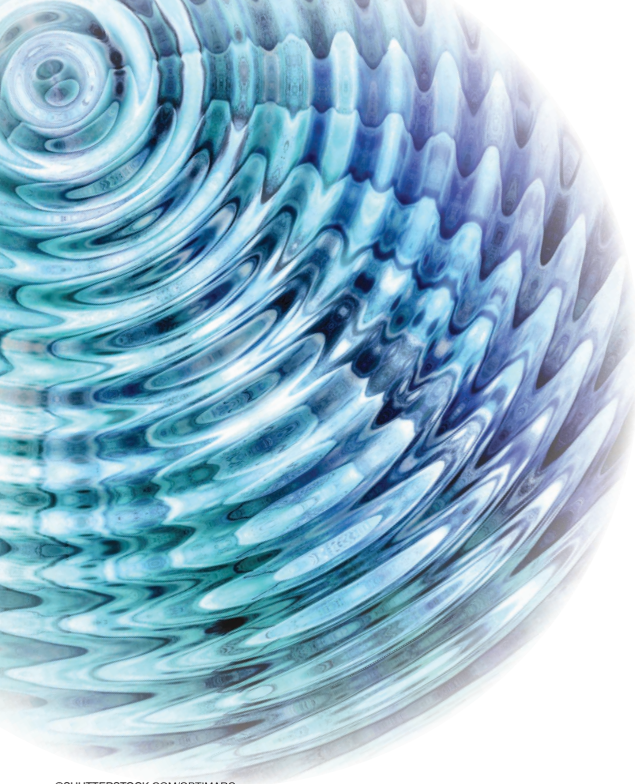




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*From theory
to applications.*

Digital Coding Metasurfaces

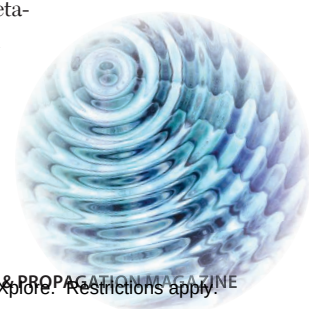
The concept of “coding metasurfaces” was first put forward in 2014 and has developed rapidly during the past few years, originating several intriguing ramifications. The essential underlying idea is to discretize the local geometry, constitutive parameters, and electromagnetic (EM) responses (e.g., the phase, amplitude, and polarization) so that they can be represented via sequences of digits (e.g., “0” and “1”). This concept breaks the boundaries between traditional analog and digital devices, bridging the physical and information worlds, and opens up new perspectives in the design of metasurfaces.

Within this framework, a broad variety of field manipulations and functionalities have been explored, including beam tailoring, holograms, and amplitude as well as polarization modulations. Remarkably, the judicious integration of switching elements (e.g., diodes) may render these platforms “programmable,” and, via dynamic modulation, may grant access to sophisticated space–time-coding schemes, which enable joint spatial–spectral field manipulations. Furthermore, by integrating sensors and artificial intelligence (AI) algorithms, self-adaptive and smart metasurfaces can be conceived for automatic sensing and reactive functionalities. In this article, we attempt a review of the developing path of coding metasurfaces, with a special emphasis on some selected research directions. Finally, we discuss possible perspectives, visions, and potential evolutions of this research area.

INTRODUCTION

Metamaterials are artificial structures that can be engineered to attain properties that are not available in natural materials, such as a negative refractive index [1]. During the past two decades, this research area has experienced sustained growth, leading to the demonstration of diverse exotic EM phenomena, such as negative refraction [2], superlensing [3], dielectric lenses [4], and cloaking and illusion effects [5]–[7]. Their 2D versions, commonly referred to as *metasurfaces*, effectively mitigate the effects of losses and fabrication challenges, thus providing technologically viable and efficient implementations [8]–[10]. In particular, generalized Snell’s laws have opened up new directions in their design [11] and have triggered diverse applications, including beam manipulations [11], vortex beams [12]–[14], holograms [15], [16], and Huygens surfaces [17]. One crucial departure from this common framework is the concept of “digital” and “coding” metastructures, which was independently introduced in 2014 by Cui et al. [18] and Della Giovampaola and Engheta [19] and opened up new pathways in the design of metamaterials and metasurfaces.

Different from the typically continuous description of materials, which can be regarded as the “analog” form, coding metamaterials or metasurfaces are represented in a digital fashion [18], [20], [21]. For example, the reflection or transmission phase response of a metasurface element can be encoded via the digits “0” and “1,” corresponding



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to 0 and 180°, respectively. Such an idea establishes a connection between the EM physics and digital information, providing a digital language to describe and design the metastructures.

Despite their discrete nature, coding metasurfaces have been proven to exhibit excellent field-manipulation capabilities. The coding idea can not only characterize the phase state, but it can be readily extended to other EM observables, such as amplitude [22], polarization [23], [24], frequency [25], and orbital angular momentum (OAM) modes [26]. Based on these concepts, a wide variety of coding metasurfaces with advanced functionalities have been proposed, including multibeam modulation [27], [28], holography [26], [29], [30], OAM beam coding [31], and full-space field manipulations [32], [33]. Interestingly, these results have been extended to other physical domains, such as acoustic [34] and thermodynamics [35]. Remarkably, the digital coding representation also enables a natural extension of information-theoretic concepts to metasurfaces, and a series of concepts and tools has been proposed [36]–[38] to quantify and control the information content in architectures based on coding metasurfaces.

A pivotal development of the coding idea is the integration of diodes or other active components, which enables the realization of multiple discrete states via reconfigurable elements. This endows a digital coding metasurface with a (re)programmable character that can be extended from the original phase states to amplitude [22], polarization [24], and reciprocity control [39]. Importantly, the reprogrammable character also enables a dynamic modulation in time, which paves the way for space-time and harmonic coding schemes for joint spatial-spectral manipulations [40], including multibeam [41], vortex beam [40], and nonreciprocal effects [42]. This has also led to the demonstration of communication prototypes working with amplitude or frequency modulations [43]–[47].

In imaging applications, programmable metasurfaces can implement different holograms and multiple illumination patterns to obtain sufficient data for reconstructing the targets [48]. Moreover, as an alternative to diode-based schemes, control mechanisms based on visible light have been demonstrated [49]. One of the most fascinating developments is the demonstration of self-adaptive smart metasurfaces integrated with sensors [50], [51], which can autonomously sense environment changes and adjust the coding patterns according to a given algorithm.

In this article, we provide a general overview of the main results, challenges, and perspectives in this fascinating and fast-paced research field, starting from the basic concepts and operation methods of coding and programmable metasurfaces. Subsequently, we focus on theoretical aspects dealing

Different from the typically continuous description of materials, which can be regarded as the “analog” form, coding metamaterials or metasurfaces are represented in a digital fashion.

with physical modeling, information, and design. Finally, we review some recent results in the areas of time-varying, multifunctional, self-adaptive, and intelligent platforms.

DIGITAL CODING METASURFACES

The general concepts of digital coding and programmable metasurfaces were first put forward by Cui et al. [18]. For the simplest binary (1-b) scenario, as shown in Figure 1(a), the structure comprises a 2D arrangement of two types of elements characterized by a reflection phase of 0 and 180°, associated with the binary digits “0” and “1,” respectively. A possible metasurface realization, based on a subwavelength metallic

patch-type unit cell, is shown in Figure 1(b). By optimizing the patch side length w , it is possible to synthesize two responses with a 180° phase difference at a specific operational frequency. Figure 1(b) shows the phase response from 7 to 14 GHz of the coding elements “0” ($w = 4.8$ mm) and “1” ($w = 3.75$ mm), with the phase difference displayed by the red-dashed curve.

Thus, considering a metasurface made of 8×8 coding elements, each possible realization can be associated to a specific “0/1” coding pattern, and the design process is greatly simplified while retaining sufficient versatility. For instance, Figure 1(c) and (d) shows two distinctly different scattering patterns (dual and quad beam, respectively) that can be realized via two coding patterns (column code “0101” and chessboard pattern “0101,” respectively). In this way, the scattered field can be shaped by suitably synthesizing the coding patterns.

It is worth highlighting that the digital coding metasurfaces described previously are passive structures. Once the design and fabrication are carried out, the corresponding EM responses and functionalities are fixed. To substantially extend the flexibility of these platforms, the concept of programmable metasurfaces was also put forward. In this case, active elements, such as varactor or positive-intrinsic-negative (p-i-n) diodes, are embedded in the unit cells to reconfigure their responses, controlled by a field-programmable gate array (FPGA), as shown in Figure 1(e). The metasurface is composed of three layers and two symmetric metallic structures, which are connected by a p-i-n diode (Skyworks, SMP-I320), are printed on the top surface of an F4B substrate [Figure 1(f) and (g)]. At 8.6 GHz, the reflection phase difference between the two states is exactly 180°. As shown in Figure 1(i), by using an FPGA to control the diodes and designing different coding sequences (such as “000000,” “111111,” “010101,” and “001011”), it is possible to obtain different scattering patterns in a flexible and programmable fashion.

BASIC THEORETICAL ASPECTS

We now move on to introducing some basic theoretical aspects underlying the modeling and information-theoretic properties

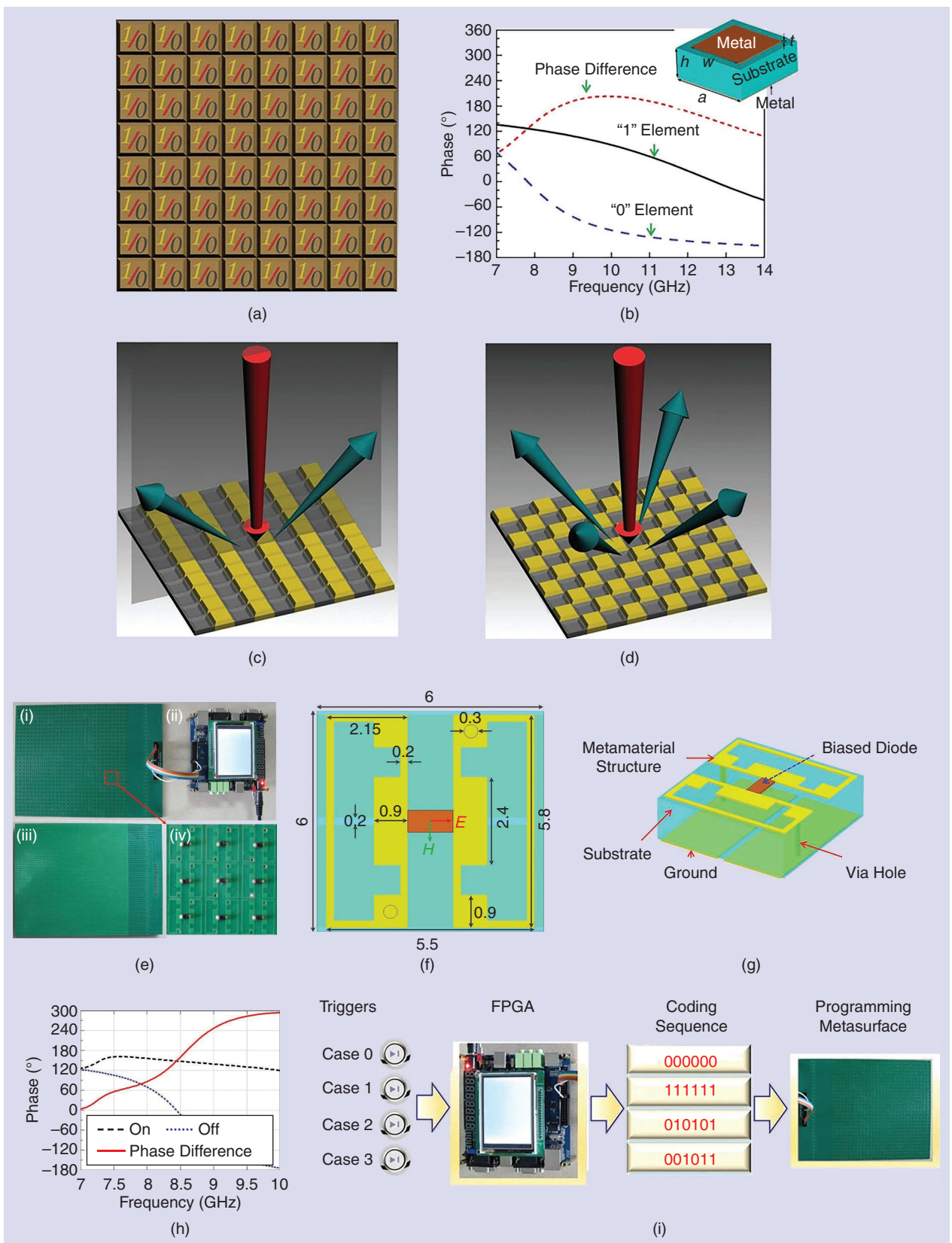


FIGURE 1. 1-b coding and programmable metasurfaces. (a) A 1-bit digital coding pattern. (b) An example of a typical coding unit cell to realize the "0" and "1" elements and the corresponding phase responses as a function of the frequency. (c), (d) Schematics of the column code "0101" and chessboard pattern "0101," implementing dual- and quad-beam responses, respectively. (e) A fabricated sample of a programmable metasurface. (f), (g) Details and a schematic of a programmable element relying on a biased diode. (h) Corresponding phase responses pertaining to the diode states. (i) Operation process of a programmable metasurface. FPGA: field-programmable gate array. (Source: [18]; used with permission.)

of digital coding metasurfaces that are essential for understanding and designing them.

SEMIANALYTICAL MODELING

Wan et al. [52] proposed an approximate modeling of coding metasurfaces that is amenable to a semianalytic treatment. In the binary case, the bits in the coding pattern are typically associated with two different reflection phases, though the magnitude and polarization may be possibly considered as well. Figure 2 shows a conceptual illustration of a coding metasurface, where the entire structure is composed of $M \times N$ equal-sized subregions with dimensions D_x and D_y along the x - and y - axes, respectively. Yellow and blue elements represent the states “0” and “1”, respectively, whose spatial distribution can be arbitrary in principle.

Under time-harmonic ($e^{j\omega t}$), normally incident, plane-wave illumination, the far field scattered by the metasurface can be approximately expressed as

$$F(\theta, \varphi) = \sum_{m=1}^M \sum_{n=1}^N \exp(j\varphi_{nm}) \times \exp\left\{jk_0 \sin\theta \left[D_x \left(m - \frac{1}{2}\right) \cos\varphi + D_y \left(n - \frac{1}{2}\right) \sin\varphi\right]\right\}, \quad (1)$$

where $k_0 = \omega/c = 2\pi/\lambda$ is the free-space wavenumber (with c and λ denoting the corresponding wave speed and wavelength, respectively), φ_{nm} is the reflection phase response (either π or zero, assuming an ideal unit amplitude) of the (m,n) th element, and θ and φ are the conventional spherical angles. This model essentially assumes that an ideal binary reflection phase can be realized and, essentially, neglects the interelement coupling. These assumptions are usually satisfied when the subregions associated with the binary digits contain a suitable number of identical subwavelength unit cells so that a local periodicity is established.

INFORMATION-THEORETIC ASPECTS

A typical scenario of interest in information theory is the transmission of information by a number of transmitters to a number of receivers through a number of channels. As shown in Figure 3(a), a pair of horn antennas (transmitting and receiving) and a metasurface can, thus, be interpreted as an information system. Inspired by the concept of Shannon information entropy, which quantifies the information carried in each message, Cui et al. [36] proposed a similar metric to quantify the information associated with a coding metasurface.

As shown in Figure 3(b), a 1-b coding provides a simple and yet general scenario to quantitatively investigate the information entropy of a metasurface, where only four combinations, namely, $G(0, 0)$, $G(0, 1)$, $G(1, 0)$, and $G(1, 1)$, can be selected in adjacent coding elements. The 2D information entropy is determined by the probability of these four selections. According to the definition of normalized Shannon entropy, a geometrical information entropy (H_1) related to the coding pattern was proposed, and a physical information entropy (H_2) was further

proposed to estimate the information carried by the far-field pattern via a Fourier transform (FT) of the coding pattern. The relationship between the geometrical (coding pattern) and physical (far-field pattern) information entropies as a function of the number of iterations is shown in Figure 3(c). It can be observed that the physical information entropy progressively increases, eventually approaching the geometrical one. This demonstrates that the physical information entropy can be controlled by the diffusion depth of the random coding pattern.

CONVOLUTION-BASED OPERATIONS

To steer a scattering pattern toward an arbitrarily predesigned direction, convolution-type operations can be exploited [37]. In signal processing, as illustrated in Figure 4(g)–(i), convolution implies a shift in the spectrum, viz.,

$$f(t)e^{j\omega t} \xrightarrow{\text{FT}} F(\omega) * \delta(\omega - \omega_0) = F(\omega - \omega_0), \quad (2)$$

with $\delta(\cdot)$ denoting a Dirac delta impulse function. Likewise, we can define the principle of scattering-pattern shift as follows:

$$E(x_\lambda)e^{jx_\lambda \sin\theta_0} \xrightarrow{\text{FT}} E(\sin\theta) * \delta(\sin\theta - \sin\theta_0) = E(\sin\theta - \sin\theta_0), \quad (3)$$

where $e^{jx_\lambda \sin\theta_0}$ denotes the electric-field distribution generated by a gradient phase pattern. Hence, by multiplying the coding pattern $E(x_\lambda)$ by the gradient coding sequence associated with $e^{jx_\lambda \sin\theta_0}$, a steering of a quantity $\sin\theta_0$ (in angular coordinates) is obtained. For illustration, as shown in Figure 4(a)–(c), we consider a 2-b coding metasurface with the four digits “0,” “1,” “2,” and “3” (corresponding to reflection phases 0, 90, 180, and 270°) represented with different shades of gray. The coding-pattern domain shows that a cross-shaped coding pattern (consisting of digits “0” and “2”) [Figure 4(a)] is multiplied by a gradient coding sequence. The result in Figure 4(c) clearly shows that, after the convolution operation with the single-beam scattering pattern, the scattering pattern [Figure 4(d)] is steered from the

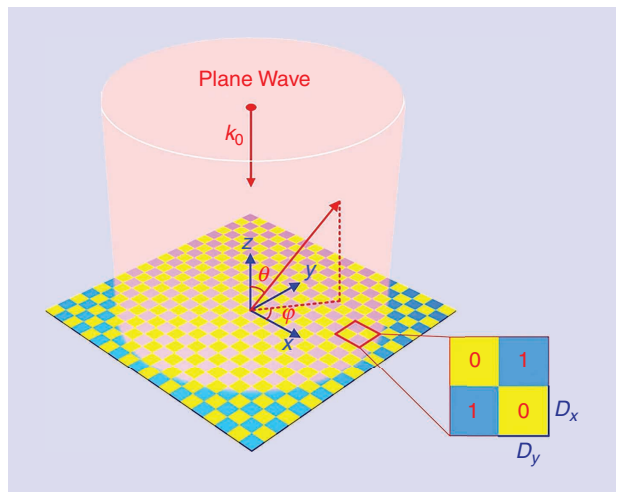


FIGURE 2. A schematic of plane-wave illumination for a coding metasurface. (Source: [50]; used with permission.)

original direction, with no apparent distortion. This is analogous to the convolution in the frequency domain schematized in Figure 4(g)–(i).

TIME-VARYING DIGITAL CODING METASURFACES

Conventional digital coding metasurfaces rely solely on spatial modulations, which, in programmable structures, can be statically changed to reconfigure the functions. Recently, there has been a growing interest in dynamic modulation, which grants access to the temporal dimension, thereby endowing additional degrees of freedom in the manipulation of EM wavefronts. In this section, we review some relevant results in the emerging fields of time-varying and space–time-coding metasurfaces.

TIME-DOMAIN-CODING DIGITAL METASURFACES

Based on related concepts, Zhao et al. [45] proposed the idea of distributing the harmonic energy via temporal modulation and applied it to a new communication system. As schematized in Figure 5, in this structure, the metasurface elements are loaded with varactor diodes. By applying a time modulation

with frequency f_0 , under a monochromatic illumination E_i at frequency f_c , the reflected field can be written as

$$E_r(f) = a_0 E_i(f) + \sum_{\nu=1}^{\infty} [a_{\nu} E_i(f - \nu f_0) + a_{-\nu} E_i(f + \nu f_0)], \quad (4)$$

where a_{ν} are complex Fourier coefficients. Hence, a series of harmonic frequencies $f_c \pm \nu f_0$ are generated, and, by suitably tailoring the temporal coding sequences, the power can be distributed among the harmonic orders. As a consequence, direct modulations of the transmitted signals can be attained without the need of external mixers.

SPACE–TIME-CODING DIGITAL METASURFACES

The concept of space–time-coding digital metasurfaces was introduced by Zhang et al. [40]. As conceptually illustrated in Figure 6(a), the idea relies on a digital, programmable metasurface whose elements' responses are switched in space and time by means of an FPGA with a 50-MHz clock frequency, according to a given coding (3D matrix of red and green dots), which

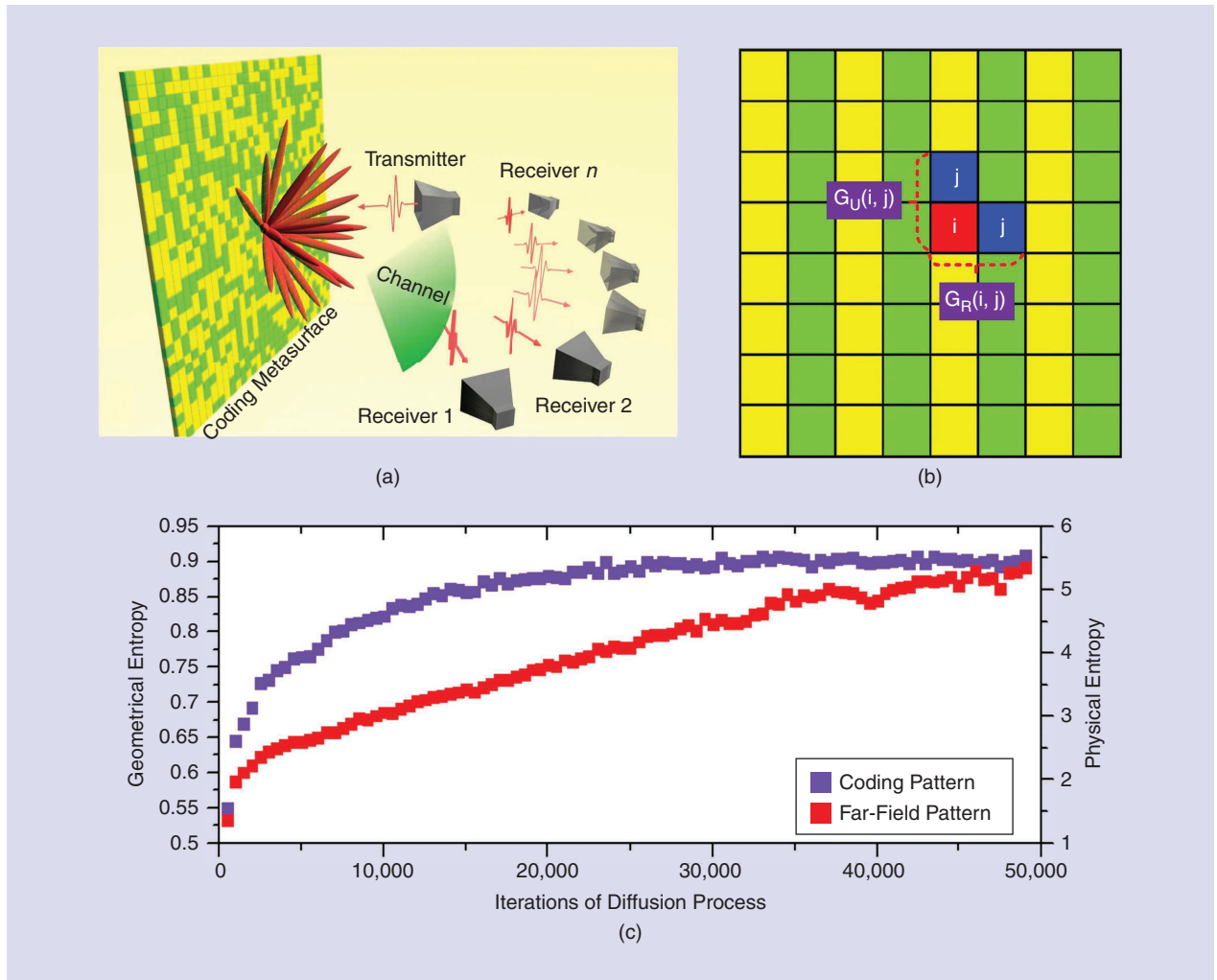


FIGURE 3. Information-theoretic aspects. (a) Schematics of possible metasurface-based information systems. (b) A conceptual calculation of the physical information entropy. (c) Geometrical and physical information as a function of iterations in the diffusion process. (Source: [36]; used with permission.)

traditional heterodyne architecture, made up of the analog-digital converter, mixer, a series of radio-frequency (RF) modules and antenna, is replaced by a programmable time-coding digital metasurface. Different from traditional wireless communication systems, the digital messages are encoded in terms of bit streams corresponding to the coding sequences implemented in the metasurface via an FPGA and transmitted to space, which greatly simplifies the system architecture. The efficiency of the proposed BFSK wireless communication system is witnessed by a measured transmitting bit rate of 78.125 kb/s.

Subsequently, Dai et al. [47] introduced multimodulation schemes for wireless communications based on time-coding digital metasurfaces. As shown in Figure 8, the metasurface unit is integrated with varactors to control the reflected phase

response. By applying suitable bias voltages, the metasurface can synthesize arbitrary constellation diagrams. By introducing a circular time shift in the control of the unit cell, this approach avoids the independent control of the amplitude and phase and successfully demonstrates different modulation schemes, namely, quadrature phase-shift keying (QPSK), eight-phase-shift keying (8PSK), and 16 quadrature amplitude modulation (16QAM).

As illustrated in Figure 8, each unit of the metasurface is integrated with two varactor diodes (Skyworks SMV-2019) and two chip capacitors (0.1 pF). A row of via holes is exploited to improve the phase range when the bias voltage changes. The largest phase range is 2.5π when the voltage varies from 0 to 21 V. The maximum amplitude fluctuation is 3.5 dB. The

metasurface is excited at the frequency of 4.25 GHz. The offset frequency is set as 100 kHz, 1 MHz, 2 MHz, and 2.5 MHz. Hence, using a time-coding digital metasurface, wireless communication systems have been demonstrated with a simplified architecture and the QPSK, 8PSK, and 16QAM modulation schemes, in which the transmission bit rate can be as high as 10 Mb/s (16QAM).

More recently, a new coding strategy was proposed by Chen et al. [46] to realize accurate and ultrabroadband manipulation of the harmonic waves. Multiple harmonic beams are generated when a monochromatic plane wave with frequency f_c illuminates the time-coding digital metasurface with a time-varying coding period $T_0 = 1/f_0$. Here, the proposed 1-b digital metasurface works at an ultrabroadband from 22 to 33 GHz with a relatively stable phase

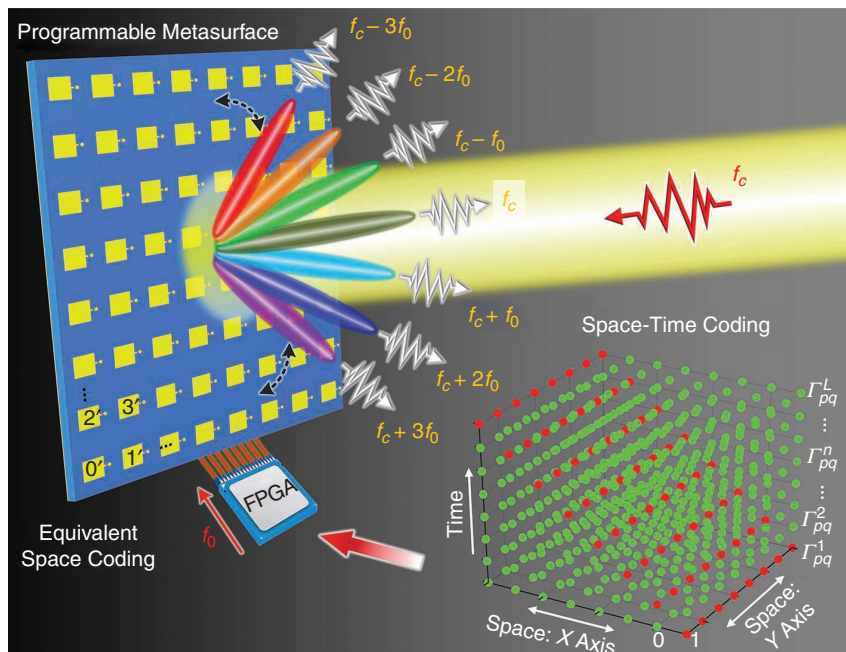


FIGURE 6. Space-time-coding metasurfaces. (Source: [40]; used with permission.)

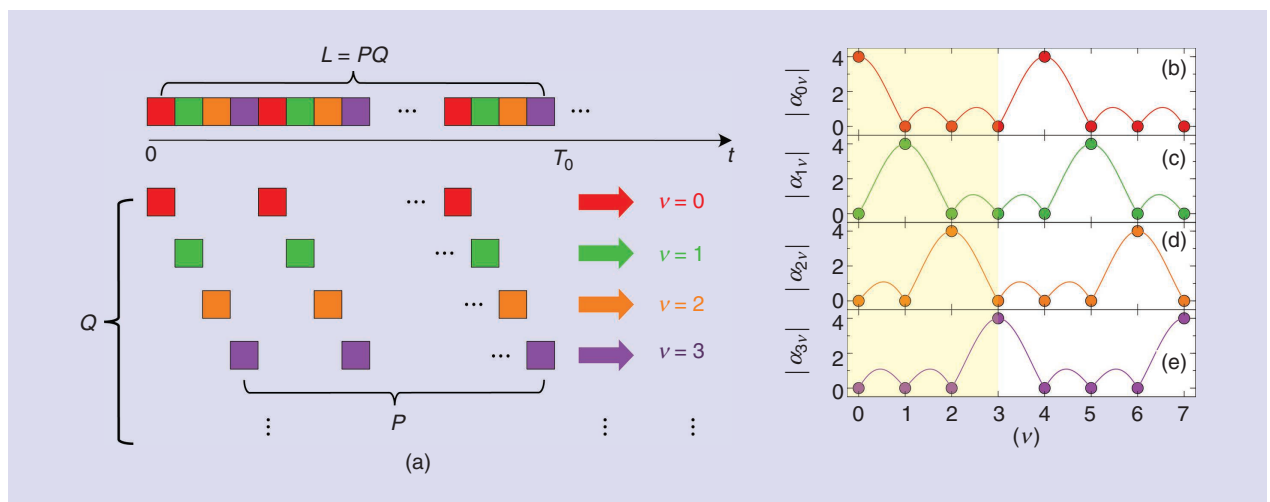


FIGURE 7. Joint multifrequency syntheses. (a) A conceptual illustration of the temporal coding scheme. (b)–(e) Magnitude responses of digital filter terms. (Source: [53]; used with permission.)

difference of 180° . By relying on the excellent performance in manipulating the EM waves using the new modulation method, a 256QAM wireless communication system was demonstrated, operating at different frequencies and different harmonics [46], as illustrated in Figure 9.

Here, the transmitted picture is first processed into a series of bitstreams, which are then mapped onto the corresponding coding sequences of the time-coding digital metasurface with a 256QAM modulation scheme at the transmitting terminal, and, hence, the information is embedded into the EM waves. Subsequently, the transmitted information can be accurately recovered by the receiver, where the received EM waves carrying the digital information are demodulated and demapped. The proposed 256QAM millimeter-wave wireless communication system has great potentials in future wireless networks due to the accurate and broadband manipulations of EM waves enabled by the new coding strategy.

Based on the powerful capabilities of a space–time-coding digital

metasurface in manipulating the frequency harmonics and spatial beams simultaneously, a new space- and frequency-division multiplexing wireless communication system was demonstrated [44]. By applying suitable time-varying coding patterns, the incident wave is scattered at different angles as different harmonic modes. In the example shown in [44], two users

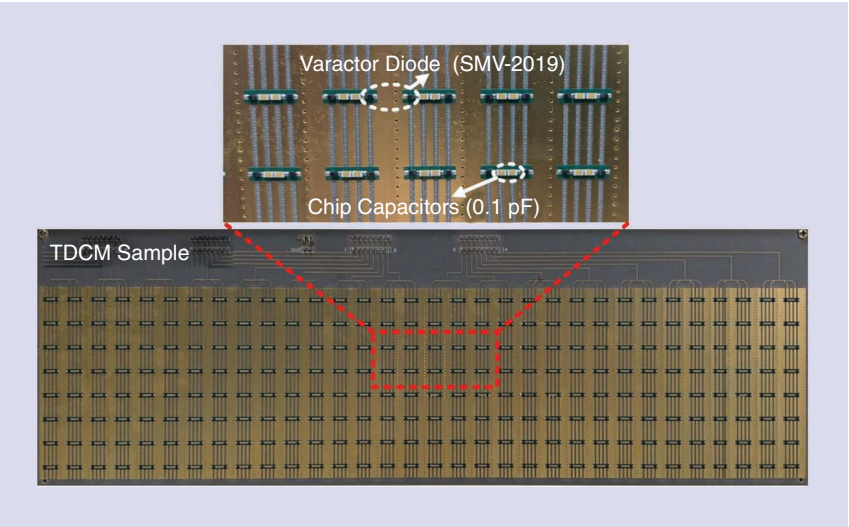


FIGURE 8. A fabricated prototype of a time-coding digital metasurface. (Source: [47]; used with permission.) TDCM: time-domain digital coding metasurface.

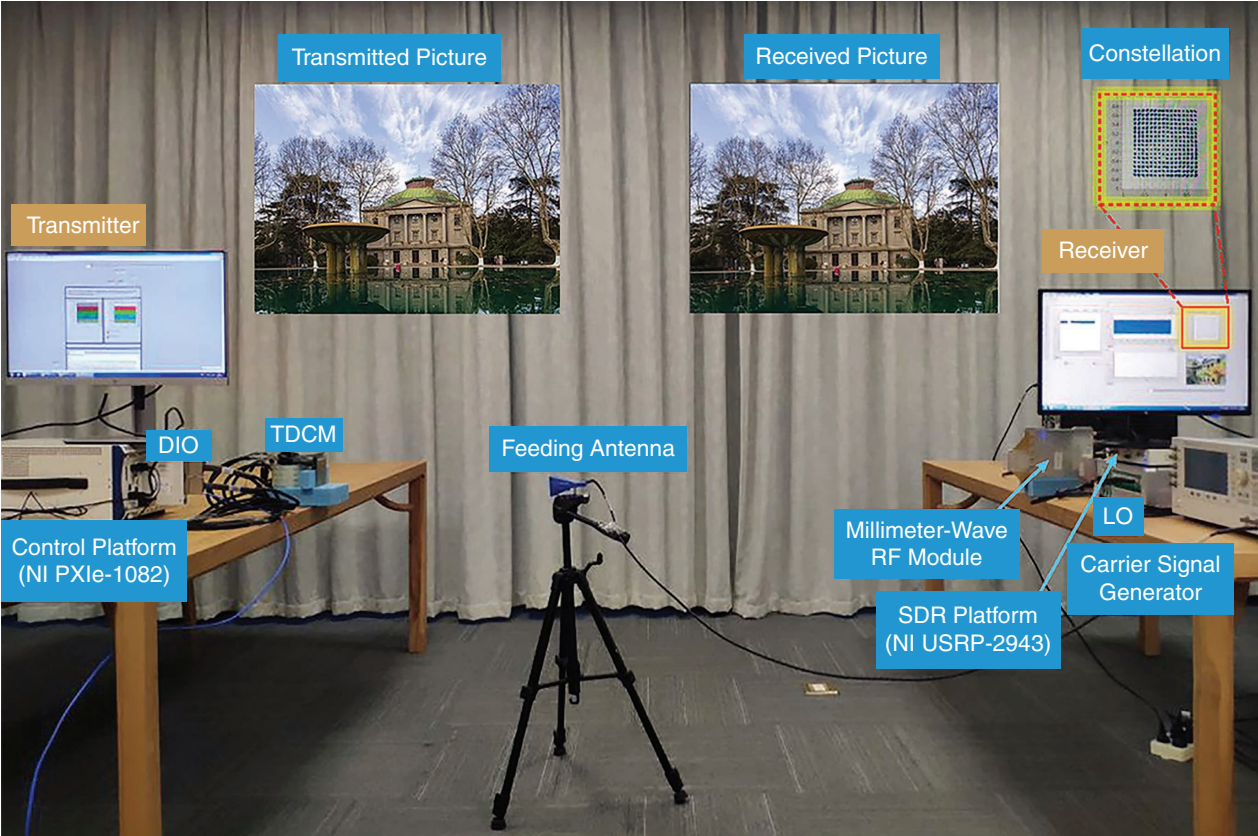


FIGURE 9. A wireless communication system based on 256QAM modulation. DIO: digital input and output; LO: local oscillator; SDR: software-defined receiver. (Source: [46]; used with permission.)

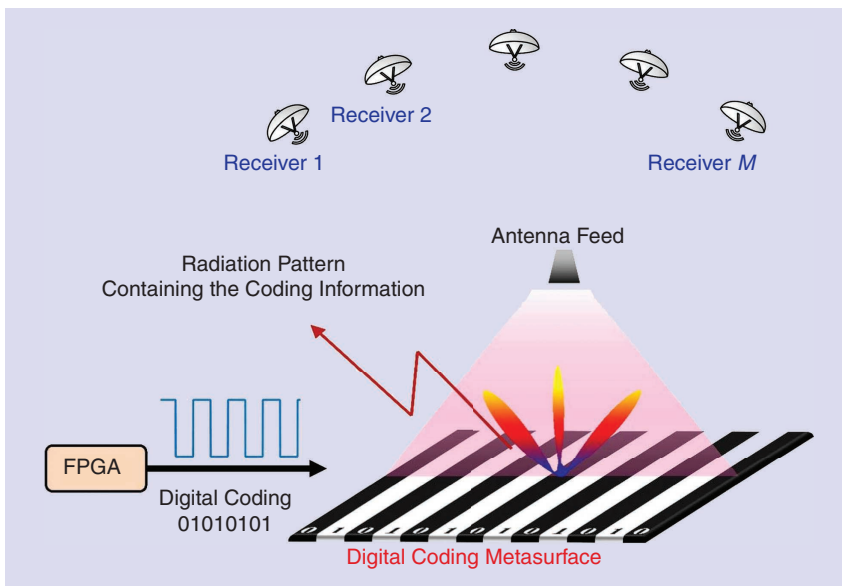


FIGURE 10. Direct communication based on programmable metasurfaces. (Source: [43]; used with permission.)

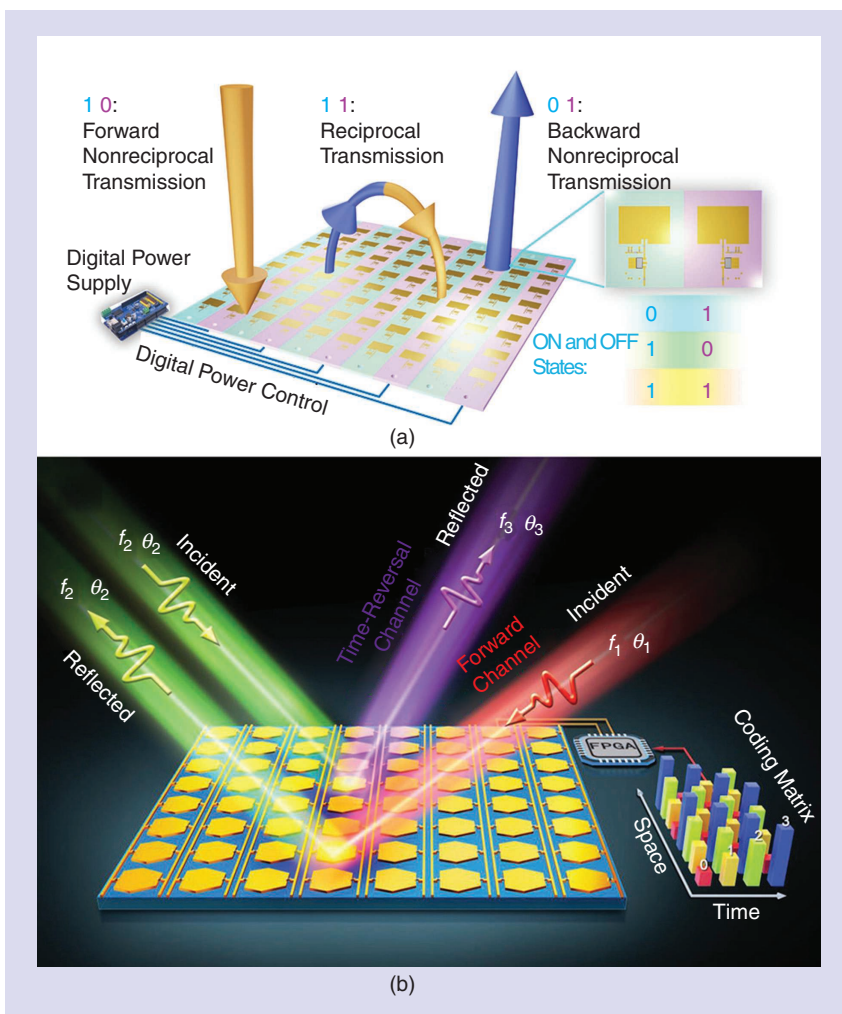


FIGURE 11. Nonreciprocal coding metasurfaces. Conceptual illustrations of programmable nonreciprocity attained via (a) PA-based and (b) space-time-coding metasurfaces. (Source: [39] and [42]; used with permission.)

at $\pm 34^\circ$ directions receive and recover two different pictures, separately and simultaneously, according to pre-designed decoding rules. The transmitter is composed of a control module, a signal generator, a horn antenna, and the metasurface. The receiving part of the user contains a horn, a down-converter, and a software-defined radio receiver (NI USRP-2943R). In this wireless system, the carrier frequency is 9.5 GHz, the diode-switching speed is 80 MHz (which achieves a modulation frequency of 10 MHz), and the measured transmission rate is 2.5 Mb/s.

MULTIFUNCTIONAL DIGITAL CODING METASURFACES

In this section, we explore five advanced functionalities of the digital coding metasurfaces, namely: direct-communication, full-space coding; nonreciprocal effects; light-driven platforms; and multiple-encrypted holography.

DIRECT COMMUNICATIONS BASED ON PROGRAMMABLE METASURFACES

A metasurface-based direct-communication system focused on spatial energy redistribution was demonstrated by Cui et al. [43], as illustrated in Figure 10. Similar to the previously illustrated time-modulated communication systems, this configuration does not require complex modulation and demodulation modules, such as a mixer, a filter, converters, and so on, and simply comprises a metasurface, an FPGA, and a horn antenna. The digital signals are directly modulated by unique coding sequences impressed to the metasurface via the FPGA by exploiting a suitably designed channel-optimization algorithm. Then, the digital information is radiated in free space via specific far-field patterns by the metasurface under the horn antenna illumination and is received by a number of antennas located at different positions in the far-field region. It is worth mentioning that the transmitted information can be correctly reconstructed from the shape of the far-field pattern only via the joint processing of several receivers, which enables significant security in the communication.

NONRECIPROCAL DIGITAL CODING METASURFACES

Engineering nonreciprocal effects in a magnetless fashion is highly desirable in antenna applications and communication systems. Recently, two nonreciprocal coding metasurfaces based on different strategies have been proposed [39], [42]. Specifically, Ma et al. [39] put forward a nonreciprocal structure composed of a coding metasurface loaded with power amplifiers (PAs). Such a transmissive metasurface, conceptually illustrated in Figure 11(a), is designed to realize editable nonreciprocity in dual directions and relies on a unit cell featuring two PAs, which can be either enabled (digital state “1”) or disabled (digital state “0”). Hence, the nonreciprocity of the proposed metasurface can be edited, and a programmable unidirectional nonreciprocity can be attained.

The second example nonreciprocal platform relies on breaking the time-reversal symmetry via space–time coding [42], as shown in Figure 11(b). To this aim, a 2-b space–time-coding metasurface composed of 16 space-coding particles is exploited, featuring two p-i-n diodes to realize four different reflection phase responses with jumps of 90° , and a space–time phase gradient is impressed. For plane-wave illumination at frequency f_1 and incidence angle θ_1 , an anomalous reflection is attained at frequency f_2 and angle θ_2 . For a nonreciprocal reflection

Multiple EM functionalities can be achieved by applying different spatial patterns to the light illuminating the optical interrogation network.

effect, a time-reversed incident wave at frequency f_2 and angle θ_1 is reflected at frequency $f_3 \neq f_1$ and angle $\theta_3 \neq \theta_1$, thereby attaining free-space isolation.

LIGHT-DRIVEN DIGITAL CODING METASURFACES

A light-driven platform was demonstrated by Zhang et al. [49] to improve the operational bandwidth and reconfigurability. As schematically illustrated in Figure 12, the system comprises 6×6 subarrays,

each composed of light-addressable metasurface elements with varactors working from 5.2 to 7.6 GHz (with 37.5% relative bandwidth) on the front and an optical interrogation network with multiple photodiodes on the back. Multiple EM functionalities can be achieved by applying different spatial patterns to the light illuminating the optical interrogation network. For instance, as shown in Figure 12, an external cloak can be implemented, where the reflected wavefront is compensated for by the light-driven metasurface as if there were no object placed on the metasurface. Moreover, “illusion” functionalities can be achieved by synthesizing the desired scattering patterns that mimic specific scattering responses. In addition, different types of OAM beams can be easily generated in real time. These schemes may set the stage for the development of novel electronic–photonic devices.

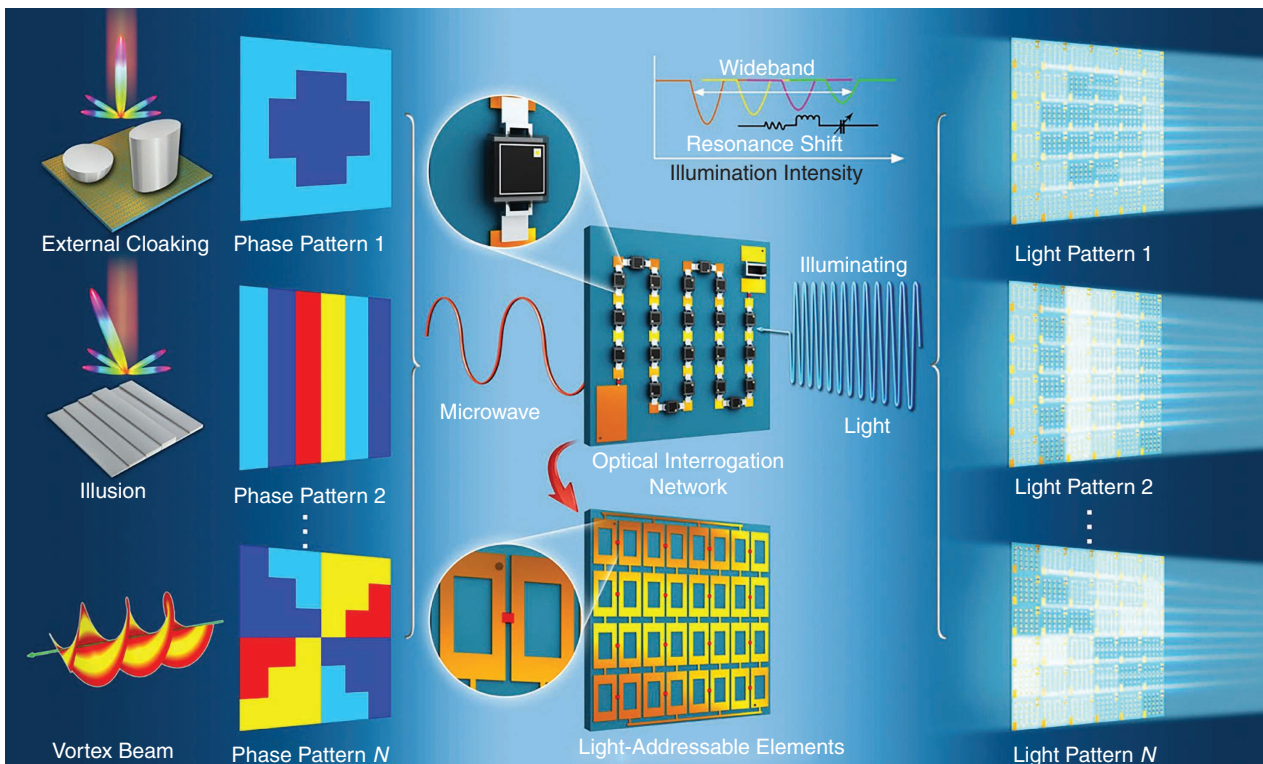


FIGURE 12. Light-driven coding metasurfaces. (Source: [47]; used with permission.)

MULTIPLE-ENCRYPTED HOLOGRAPHY

By relying on the high information capacity and security of OAM beams, with a great number of mutually orthogonal helical modes, Xiao et al. [26] proposed an OAM-encrypted holographical system based on coding metasurfaces. As schematized in Figure 13, the system is composed of two coding metasurfaces, one of which generates the OAM beams, while the other one implements the encryption by superposing different OAM helical-phase wavefronts on the transmitted images. As an example (Figure 13), two different images (“X” and “Y”) stored are correctly decrypted only when the second metasurface is illuminated by an OAM beam with proper helicity. Otherwise, the transmitted information will be noisy and unreadable, thereby rendering the system inherently secure.

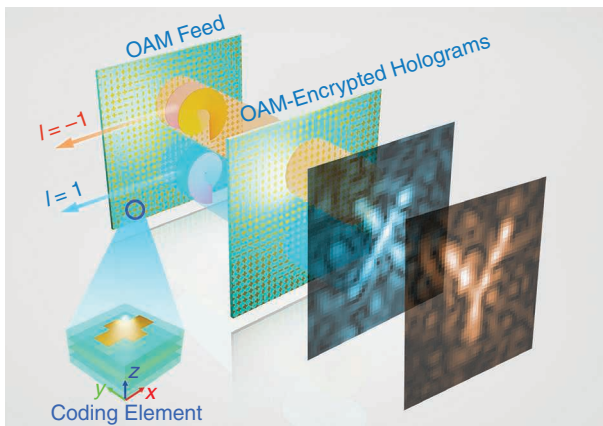


FIGURE 13. Multiple-encrypted EM holography. (Source: [26]; used with permission.)

SELF-ADAPTIVE AND INTELLIGENT DIGITAL CODING METASURFACES

In all scenarios discussed so far, the main focus was on field manipulations in static or dynamic fashions. However, the programmable character of digital coding metasurfaces can be fruitfully exploited to attain smart functionalities in many applications, such as sensing and imaging. In this section, we review three scenarios, including self-adaptive configurations, smart sensing, and intelligent imagers.

SELF-ADAPTIVE METASURFACES

The concept of smart and intelligent coding metasurfaces, conceptually illustrated in Figure 14(a), was proposed by Ma et al. [50]. Based on a programmable metasurface, sensors and feedback components are introduced to form a closed loop that allows switching the functionalities without the need for manual commands. The sensors integrated into the metasurface are designed to detect changes in the metasurface and its surrounding environment (e.g., the spatial posture, state of movement, temperature, and so on) and send the information in digital form to the microcontroller unit (MCU). The latter, via preloaded feedback algorithms, can adaptively change the coding patterns of the metasurface in response to changes in these values by giving real-time commands to the FPGA.

As a proof-of-concept example, a satellite communications scenario with an airplane is considered in Figure 14(a). As illustrated in Figure 14(b), by integrating gyroscopic sensors in the unit, as the airplane changes its spatial position in the sky, it is possible to detect the rotation angle of the metasurface and send the corresponding information to the MCU

every half second. The MCU, in turns, processes the data and executes a fast inverse-design algorithm to adjust the desired coding pattern in real time so that the primary beam reradiated by the metasurface always points toward the satellite irrespective of its rotation. To rapidly obtain the desired coding pattern, an inverse-design algorithm is developed to calculate the pattern with beam-scanning angles from 5 to 85°, in which the step accuracy is 1°.

SMART SENSING METASURFACES

Most sensing metasurfaces are designed to sense and manipulate different objects, which limits their application prospects. Ma et al. [51] proposed a smart sensing metasurface with self-defined functions to simultaneously achieve the sensing and manipulation of microwave scattering fields. As schematized in Figure 15, the platform is an evolution of a phase-programmable metasurface, featuring sensing units and executing units.

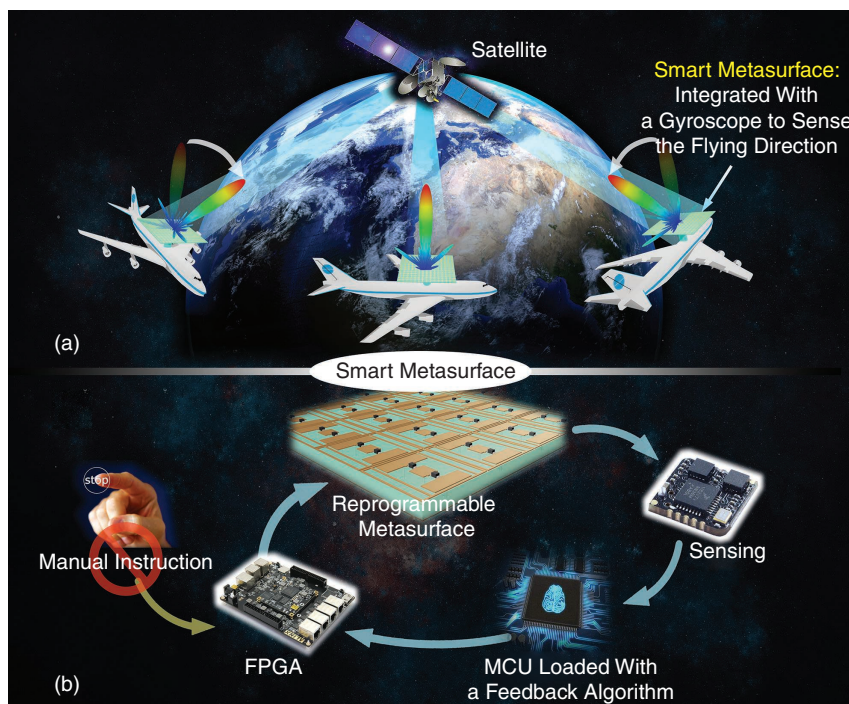


FIGURE 14. Smart metasurfaces. Conceptual illustrations of (a) self-adaptive metasurfaces and (b) a closed-loop system. MCU: microcontroller unit. (Source: [48]; used with permission.)

The sensing units are able to recognize the power level of impinging waves with different polarizations and transmit the data to the MCU for digitization. After the FPGA collects the data, it can generate different coding patterns for incident waves with different polarizations based on a preloaded algorithm. Each sensing unit is equipped with a sensing module, which can feed back the sensing voltage to the FPGA. When the power level of the signal generator exceeds 10 dBm, the output voltage increases significantly from 0.15 to 0.3 V, which can be sensed by the MCU. Therefore, it is possible to implement a smart sensing metasurface with self-defined functions in dual polarizations.

A metasurface containing 10×10 unit cells was fabricated with four distributed sensing units. The operating speed is about $15.2 \mu\text{s}$, from sensing the incidence to the coding pattern deployed by the FPGA. As shown in Figure 15, two coding patterns (A and B, represented by different colors) are arranged synchronously along the x - and y -directions on the metasurface. Under different polarizations of the illuminated waves, the metasurface will respond with dual-beam scattering patterns directed along orthogonal (x and y) directions.

INTELLIGENT IMAGING

The applications of intelligent programmable metasurfaces are not limited to field manipulations and, for instance, can be extended to imaging and recognition problems. Within this framework, Li et al. [54] proposed an intelligent metasurface imager driven by artificial neural networks (ANNs) that is capable of collecting the EM data from the target adaptively and performing the target imaging and recognition. By comparison with traditional imaging systems, this platform does not need any complex hardware.

As illustrated in Figure 16(a), the intelligent metasurface can detect human activities in rooms and, thanks to its low profile, can be easily mounted on walls. The metasurface is composed of 32×24 elements with independent phase control via a p-i-n diode (SMP1345-079LF). Figure 16(b) illustrates the system architecture, which can operate either actively (by emitting a signal and collecting the scattered data) or passively

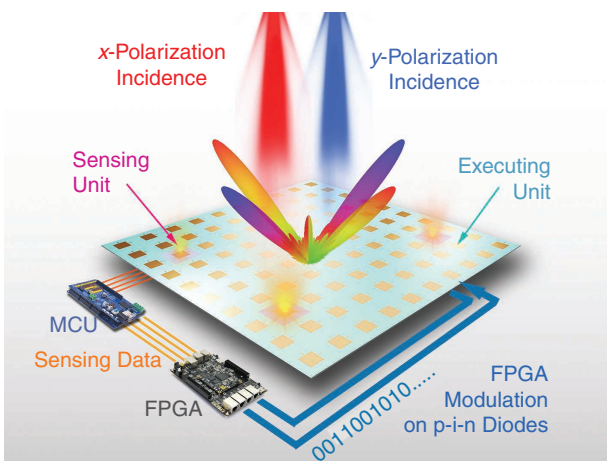


FIGURE 15. A smart sensing metasurface. (Source: [49]; used with permission.)

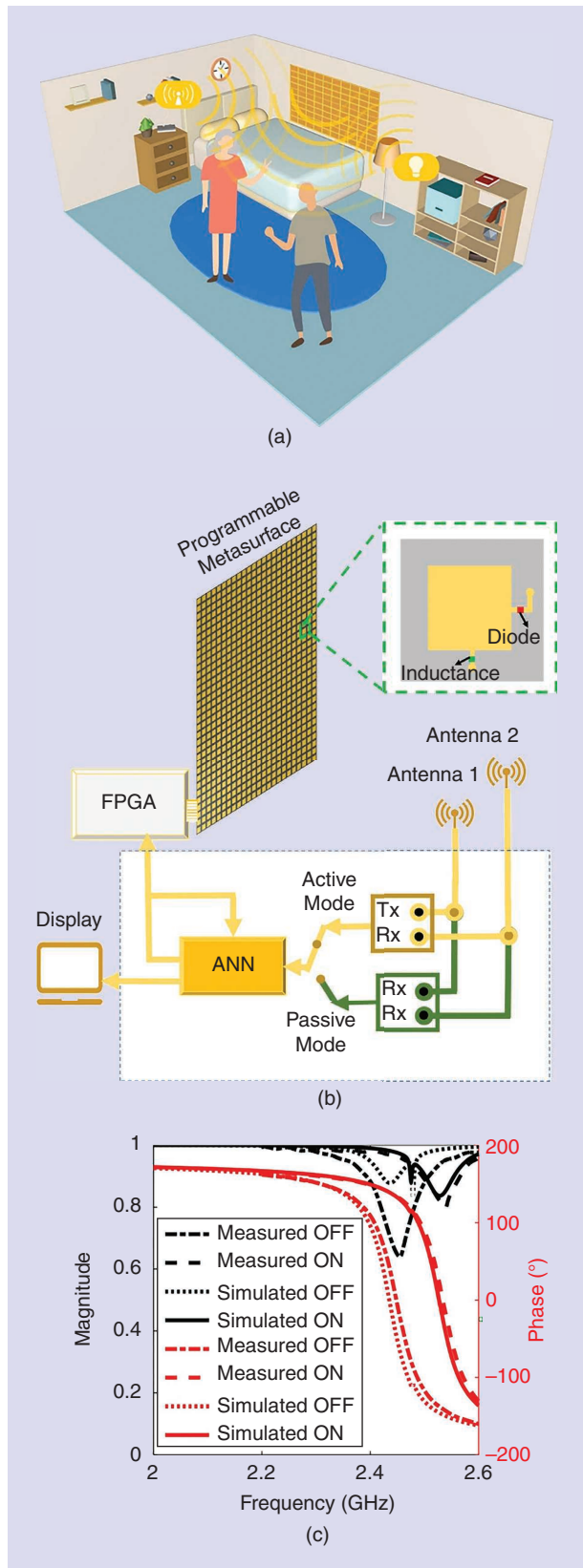


FIGURE 16. An intelligent imager based on a programmable metasurface. (a) An assumed scenario of application. (b) The architecture of the imaging and recognition system. (c) The amplitude and phase responses of the programmable element. Rx: receive; Tx: transmit. (Source: [52]; used with permission.)

(by exploiting stray signals from Wi-Fi sources). In both the active and passive modes, the metasurface switches 53 coding patterns to collect scattering data from the environment. As shown in Figure 16(c), the 1-b programmable unit was designed and validated so as to generate a 180° phase change at about 2.45 GHz, where the reflected amplitude is fairly strong.

CONCLUSIONS

In summary, we have reviewed the basic theoretical aspects of coding metasurfaces and some selected results in recent years, from the early passive configurations to various active and programmable designs. These results clearly illustrate the capabilities and potentials of these platforms and establish a fascinating connection between the physical and digital domains. From the development process, we summarize the trajectory of coding metasurfaces in three main stages:

- the initial exploration of passive coding metasurfaces, with basic theory and static EM-field manipulations, including beam scanning, multiple beam, vortex beam, and so on
- digitally programmable metasurfaces and corresponding programmable EM manipulations, including space–time-coding metasurfaces, full-space EM modulation, and so on
- smart configurations, including self-adaptive and intelligent sensing/imaging, which are still largely unexplored.

As possible future developments, we envision a deeper fusion among information, metasurfaces, and EM-field manipulations. One important direction is toward intelligent metasurfaces, which mainly includes two ramifications at the present stage: 1) AI-powered designs and 2) sensor-integrated programmable platforms. The next generation of intelligent metasurfaces will exhibit more advanced sensing and recognition capabilities to actively learn the EM characteristics in real-world scenarios and appropriately respond and will enable the execution of complex computing tasks directly in the EM wave space. We believe that coding and programmable metasurfaces will play a crucial role in the developments of next-generation wireless communication and radar systems.

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